



Handout 6.7: Quick Bread Lab Sheet

Name: _____ Date: _____ Period: _____

Station: _____ My role: _____

Print for every student. Bring it both days. Day 1 is pages 1 to 3, Day 2 is pages 4 and 5.

DAY 1

Part 1. How leavening works

Write it in four sentences, in your own words.

1. _____
2. _____
3. _____
4. _____

Which one needs an acid to work? Circle: baking powder / baking soda

Where does the acid come from in a real recipe? Name three sources:

1. _____
2. _____
3. _____

Part 2. The four-cup fizz test

Cup	What is in it	What happened (none, a little, a lot)
1	Baking soda + water	
2	Baking soda + vinegar	
3	Baking powder + water	
4	Baking powder + vinegar	

The rule, in one sentence:

Baking soda needs _____, and baking powder does not because

Answer: _____

Part 3. The experiment setup

The four batches

Batch	Leavening treatment	What I predict
A (control)	1 tsp baking powder	
B	none at all	
C	2 tsp baking powder	
D	1 tsp baking soda, no acid	

Variables

Independent variable (the one thing we change on purpose):

Line 1: _____

Dependent variable (what we measure to see the result). Name all three. Hint: look at the columns in the Part 5 data table.

1. _____ 2. _____ 3. _____

Constants (everything we keep exactly the same). List at least four. Try for eight:

1.	5.
2.	6.
3.	7.
4.	8.

Why do we need Batch A at all, if we already know a normal muffin rises?

Answer: _____

Line 2: _____

Part 4. My hypothesis and prediction

My hypothesis (my best guess, one sentence, in the form “if we change , *then* ”):

Line 1: _____

Line 2: _____

My prediction: rank the four batches by how high I think they will rise.

Highest ____ second ____ third ____ lowest ____

Batch D prediction: what will it taste like and why?

Answer: _____

Day 1 exit card

The one thing we are changing is _____.

Batch ____ will rise the highest because _____.

DAY 2

Part 5. Data

Use the ruler. Measure the batter height in the liner before baking. Measure the muffin height after baking.

Batch	Height BEFORE baking	Height AFTER baking	How much it rose	Crumb (none, few, or many holes)	Taste
A (control)					
B (no leavening)					
C (double)					
D (soda, no acid)					

Time the tin went in: ____ **Time it came out:** ____ **Oven temperature:** ____

Actual ranking by height

Highest ____ second ____ third ____ lowest ____

Compare to your prediction

Batch	I predicted	What actually happened	Same or different?
A			
B			
C			
D			

Part 6. Conclusion

Fill in all four parts.

The data showed _____

Line 2: _____

My hypothesis was (circle one) **supported** / **not supported** / **partly supported**, because

Answer: _____



My evidence is _____

Line 3: _____

One thing that could have gone wrong with our test is _____

Line 5: _____

One sentence our station can read out loud at the station check:

Answer: _____

Station check

Line	Done?
Oven off, tin on the trivet, nothing hot left out	
All four bowls washed, dried, and put away per the drawer map	
Measuring cups and spoons washed and put away	
Counters wiped and sanitized	
Sink empty and rinsed	
Floor swept, trash out	
Leavening jars returned to the teacher's table	

Teacher signature: _____

Grade 8 stretch: Batch E

Batch E: 1 tsp baking soda PLUS 1 tsp white vinegar. Everything else the same.

Batch E	Before	After	How much it rose	Crumb	Taste

Does Batch E behave more like Batch A or more like Batch D? _____

What does that prove? _____

Line 3: _____

Name one confounding variable in our own test. (A confounding variable is something we did not control that could have changed the result.) **Then say how you would control it next time.**

The variable: _____

How to control it: _____